Why You Should Switch *to* Extendable Mesh Networking Routers

Isn't it inconvenient and time consuming to manually connect to the Wi-Fi network every time you change rooms or move to different floors in your organisation? Thankfully, with the arrival of mesh topology networking technology, Wi-Fi routers can now handle the challenge of complex routing within a large organisation, even covering an area as big as 557.4sqm (6000sq ft).

The dream of seamless connectivity will soon be a reality. While it was nearly impossible to connect an entire city with wireless technology earlier, wireless mesh networks will now enable us to connect an entire enterprise, large organisations, government bodies and college campuses, without any interruption.

Wireless mesh routers were introduced in the Indian market in January 2018, and are ready to be used at the consumer as well as the enterprise levels.

Why go for extendable routers based on mesh technology?

The frequent changes in access point authentication, the change in network connectivity speed, low bandwidth and poor connectivity are some of the challenges faced by standalone routers. And these problems are well addressed by mesh technology based extended routers.

When the area to be covered for seamless wireless connectivity is as big as 557.4m, mesh topology acts as a boon to organisations since the traditional wireless networking topology fails to deliver the required bandwidth. For example, if you are moving from one conference room on the first floor to another one on the third floor, the wireless access point changes, asking for authentication prior to being given access to the network. This often results in poor connectivity and can disrupt the ongoing meeting.

Wireless mesh nodes can be moved or removed depending on the location you are installing them in. This technology is efficient in the areas where the wireless network is intermittently blocked. For instance, an old building does not allow wireless signals to pass through the brick walls as the infrastructure is not adapted to wireless technology. Mesh wireless networking technology is adaptable and expandable, and allows the network signals to pass through the walls. Also, it provides uninterrupted service in warehouses and transportation settings.

The demand for wireless connections is high when it comes to emergency responses and public safety, applications that typically mean covering a large geographical area. A wireless mesh equips you with high speed mobility and good-quality video surveillance as it delivers high throughput and very reliable wireless connectivity. This WMN (wireless mesh network) technology is a combination of WLANs (wireless large area networks) and ad hoc networks that form an intelligent, large scale and broadband network. Telemedicine projects use WMNs in emergency room links.

Dynamic changes in network topology, scalability, self-management and self-healing are some of the major features of WMN technology. If you are looking to cover large areas without sacrificing the quality of the wireless connection, wireless mesh is the technology that will solve your connectivity issues.

The traditional standalone Wi-Fi routers give you a hard time once you lose connectivity as you move further away from the router. But a mesh network creates a continuous wireless link within each access point, establishing a connection throughout your office or home and reducing the possibility of dead zones. The MIMO (multiple input and multiple output) routers are replaced with wireless mesh routers, which can also talk to multiple devices simultaneously.